

CS556-HW3 - Vector Spaces and Subspaces

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Question 1 (10 points) Write the complete solution of the following linear system:

$$x + 2y - z = 1$$

$$3x + 5y + 2z = 3$$

$$2x + y + 13z = 2$$

Solution

$$\begin{bmatrix} 1 & 2 & -1 \\ 3 & 5 & 2 \\ 2 & 1 & 13 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

$$R2 = R2 - 3R1$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 5 \\ 2 & 1 & 13 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$R3 = R3 - 2R1$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 5 \\ 0 & -3 & 15 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$R2 = -1 R2$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -5 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ RREF}$$

Infinite solutions since the last row is all 0's.

Question 2 (10 points) Find the rank of the following matrix:

$$\begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{bmatrix}$$

Solution

Swap R1, R2

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 3 & 1 & 1 & 3 \end{bmatrix}$$

$$R3 = R3 - 3R1$$

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & -5 & -8 & 3 \end{bmatrix}$$

$$R3 = R3 + 5R2$$

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 2 & 2 \end{bmatrix}$$

$$R2 = 1/2R2$$

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Rank is 3

Question 3 (10 points) Construct a matrix A whose column space contains vectors $\begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$, and whose null

space contains the vector $\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$.

Solution

$$A = \begin{bmatrix} 3 & 4 & a \\ 6 & 0 & b \\ 2 & 1 & c \end{bmatrix}$$

$$x = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$$

Null Space, $Ax = 0$

$$6 + 8 + 1a = 0, a = -14$$

$$12 + 0 + 1b = 0, b = -12$$

$$4 + 2 + c = 0, c = -6$$

$$\text{So } A = \begin{bmatrix} 3 & 4 & -14 \\ 6 & 0 & -12 \\ 2 & 1 & -6 \end{bmatrix} \text{ is a matrix with the two column vectors and a Null Space which contains the vector } \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$$

Question 4 (10 points) Compute the following matrix-vector multiplication as:

$$\begin{bmatrix} 2 & 1 & 3 \\ 7 & 1 & 0 \\ 3 & 5 & 9 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$$

a) Linear combination of columns

$$2 \times \begin{bmatrix} 2 \\ 7 \\ 3 \end{bmatrix} + 4 \times \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} + 1 \times \begin{bmatrix} 3 \\ 0 \\ 9 \end{bmatrix} = \begin{bmatrix} 11 \\ 18 \\ 35 \end{bmatrix}$$

b) Dot product of rows

$$\begin{bmatrix} 2 & 1 & 3 \\ 7 & 1 & 0 \\ 3 & 5 & 9 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 11 \\ 18 \\ 35 \end{bmatrix}$$

Question 5 (10 points) Find the value of k for which the matrix has:

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 3 & 1 \\ 4 & 8 & k \end{bmatrix}$$

a) Dependent Columns

$$\begin{aligned} \det(A) &= 1 \begin{vmatrix} 3 & 1 \\ 8 & k \end{vmatrix} - 3 \begin{vmatrix} 2 & 1 \\ 4 & k \end{vmatrix} + 2 \begin{vmatrix} 2 & 3 \\ 4 & 8 \end{vmatrix} \\ &= (3k - 8) - 3(2k - 4) + 2(4) \\ &= 3k - 8 - 6k + 12 + 8 \\ &= -3k + 12 \\ k &= \frac{-12}{-3} = 4 \end{aligned}$$

For the columns to be dependent the determinate should be 0. Setting $k = 4$ causes the determinate to be 0.

b) Independent Columns

Choosing any number other than 4 would cause the determinate not to be 0 and therefore the columns would be independent.

Question 6 (20 points) Find a basis for the four fundamental subspaces of the matrix.

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

Null Space $N(A)$, ($Ax = 0$)

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R2 = R2 - R1$

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R3 = R3 - R2$

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ F & P & F & P & F \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Column Space $C(A)$, ($Ax = b$)

The basis for the column space is $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 6 \\ 2 \end{bmatrix}$.

Row Space $= R(A) = C^T$

The non-zero rows form the row space, $\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 & 2 & 4 & 6 \end{bmatrix}$

Null Space $= N(A)$, ($Ax = 0$)

$$\text{Let } \hat{x} = [x_1, x_2, x_3, x_4, x_5]^T$$

From Row 2: $x_4 + 2x_5 = 0$ and so $x_4 = -2x_5$

From Row 1: $x_2 + 2x_3 + 3x_4 + 4x_5 = 0$, Substituting x_4 into the equation yields

$$x_2 + 2x_3 + 3(-2x_5) + 4x_5 = 0$$

$$x_2 + 2x_3 - 6x_5 + 4x_5 = 0$$

$$x_2 + 2x_3 - 2x_5 = 0$$

$$x_2 = -2x_3 + 2x_5$$

Let $x_1 = t, x_3 = s, x_5 = r$

$$\hat{x} = \begin{bmatrix} t \\ -2s + 2r \\ s \\ -2r \\ r \end{bmatrix} = t \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + r \begin{bmatrix} 0 \\ 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

so the basis for the null space is

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

Left Null Space = $N(A)^T$

$$A^T = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 2 & 0 \\ 3 & 4 & 1 \\ 4 & 6 & 2 \end{bmatrix}$$

Let $\hat{y} = [y_1, y_2, y_3]^T$

$$0y_1 + 0y_2 + 0y_3 = 0$$

$$y_1 + y_2 = 0, \text{ or } y_1 = -y_2$$

$$2y_1 + 2y_2 + 0y_3 = 0 \text{ or } y_1 = -y_2$$

$$3y_1 + 4y_2 + 1y_3 = 0 \text{ or when substituting } -y_2 \text{ for } y_1 \quad -3y_2 + 4y_2 + y_3 = y_2 + y_3 = 0 \text{ or } y_3 = -y_2 = y_1$$

$$4y_1 + 6y_2 + 2y_3 = 0$$

Finally, if we let $y_2 = s, y_1 = -s, y_3 = -s$ we have the basis of $\begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$

Question 7 (10 points) Consider the vectors $\hat{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\hat{w} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ a) In the x-y plane mark all nine linear combinations $c\hat{v} + d\hat{w}$, with $c = \{-2, 0, 2\}$ and $d = \{0, 1, 2\}$.

$$-2\hat{v} + 0\hat{w} \quad (-2, 0)$$

$$-2\hat{v} + 1\hat{w} \quad (-2, 1)$$

$$-2\hat{v} + 2\hat{w} \quad (-2, 2)$$

$$0\hat{v} + 0\hat{w} \quad (0, 0)$$

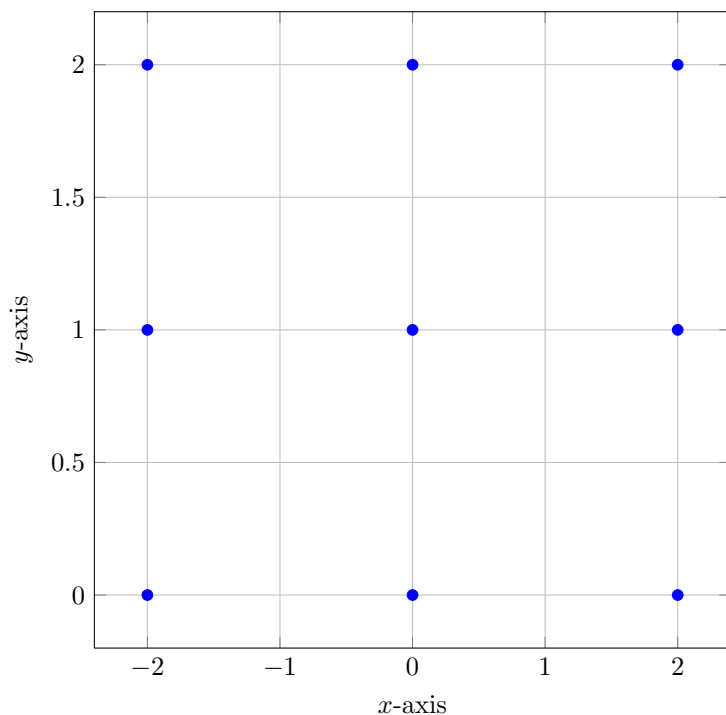
$$0\hat{v} + 1\hat{w} \quad (0, 1)$$

$$0\hat{v} + 2\hat{w} \quad (0, 2)$$

$$2\hat{v} + 0\hat{w} \quad (2, 0)$$

$$2\hat{v} + 1\hat{w} \quad (2, 1)$$

$$2\hat{v} + 2\hat{w} \quad (2, 2)$$



b) What shape do all linear combinations $c\hat{v} + d\hat{w}$ fill? A line? The whole plane? Are the vectors $\hat{v}$ and $\hat{w}$ independent?

The linear combinations span the whole plane.

From computing the determinate will can assess the independence of the vectors $\hat{v}$ and $\hat{w}$.

$$\det \begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix} = (1 \times 0 - 1 \times 2) = -2$$

The vectors are independent because the determinate is not 0.

Question 8 (10 points) Consider the vectors $\hat{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\hat{v} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$ and $\hat{w} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$.

a) Can you solve the system $x\hat{u} + y\hat{v} + z\hat{w} = \hat{b}$ if $\hat{b} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$?

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & -1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

R3 = R3 - R1

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{bmatrix}$$

R3 = R3 + R2

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

-1R2, -1R3

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$R2 = R2 + R3$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$R1 = R1 - 2R3$$

$$\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$x = 2 \quad y = -1 \quad z = -1$$

b) What if $\hat{b} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$? How many solutions are there?

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & -1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

$$R3 = R3 - R1$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & -2 & 0 \end{bmatrix}$$

$$R3 = R3 + R2$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$$-1R2, -1R2$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$R2 = R2 + R3$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$R1 = R1 - 2R3$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

There are infinitely many solutions?

c) Are the vectors $\hat{u}$, $\hat{v}$, and $\hat{w}$ dependent or independent?

d) Use parts (a) - (c) to decide if $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ is an invertible matrix or not.

Question 9 (10 points) Consider the linear system for some constants b and g :

$$x - 2y + 3z = 3$$

$$2x + y + bz = -4$$

$$x + 0y + 1z = g$$

Converted to an augmented matrix

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 2 & 1 & b & -4 \\ 1 & 0 & 1 & g \end{array} \right]$$

Gaussian Elimination

$$R2 = R2 - 2R1$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 0 & 5 & b-6 & -10 \\ 1 & 0 & 1 & g \end{array} \right]$$

$$R3 = R3 - R1$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 0 & 5 & b-6 & -10 \\ 0 & 2 & -2 & g-3 \end{array} \right]$$

$$R3 = R3 - \frac{2}{5}R2$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 0 & 5 & b-6 & -10 \\ 0 & 0 & \frac{2}{5}(1-b) & g+1 \end{array} \right]$$

a) What constant b makes the system singular (missing a pivot).

When the third row's coefficient for z is zero. This is the case when $\frac{2}{5}(1-b) = 0$, which is the case when $1-b=0$, $b=1$.

b) For the value of b found in Part (a), for which values of g , does the system have infinitely many solutions?

Continuing for $b=1$ we can see the values of g for which the system has infinitely many solutions.

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 0 & 5 & b-6 & -10 \\ 0 & 0 & 0 & g+1 \end{array} \right]$$

For there to be infinitely many solutions the third row solution should be $0=0$ which is the case when $g=-1$.

c) Find two distinct solutions of the system for that g .

For the values $b=1$ and $g=-1$ the system simplifies to

$$x - 2y + 3z = 3$$

$$2x + y + 1z = -4$$

$$x + z = -1 \text{ or } x = -1 - z$$

Substituting $x = -1 - z$ back into the other two equations yields

Substitution 1

$$(-1 - z) - 2y + 3z = 3$$

$$-2y + 2z = 4$$

$$-y + z = 2$$

$$z - y = 2$$

Substitution 2

$$2(-1 - z) + y + z = -4$$

$$y - z = -2$$

$$y = z - 2$$

so we have

$$x = -1 - z$$

$$y = z - 2$$

z is free

Solutions

For $z=0$, $x = -1 - 0 = -1$, $y = 0 - 2 = -2$

For $z = 1$, $x = -1 - 1 = -2$, $y = 1 - 2 = -1$